Course: Data Structures (CSE-1201) **Time: 1 Hour**

Any 2 from (1-4) and (5-7) compulsory. **Marks: 5*5=25**

1. Sort the items using Quick Sort algorithm: 44, 33, 11, 55, 77, 44, 40, 90 and 60.

2. Sort the following array, A= <23, 17, 14, 6, 13, 10, 1, 5, 7, 12> using heap sort. Is the array with values a max heap?

3. Write down the algorithms of Insertion Sort and analyze the best case and worst case complexity of the algorithm.

4. Write down reason of using doubly liked list instead of singly linked list. How to reverse a linked list using recursion with examples.

5. Write a function to insert an element in the middle of a Linked List. Let consider the following linked list with 12 as the head. Now write an algorithm to insert a new node (15) in the middle of node 18 and 7.

6. While we use infix expressions in our day to day lives. Computers have trouble understanding this format because they need to keep in mind rules of operator precedence and also brackets. Prefix and Postfix expressions are easier for a computer to understand and evaluate.

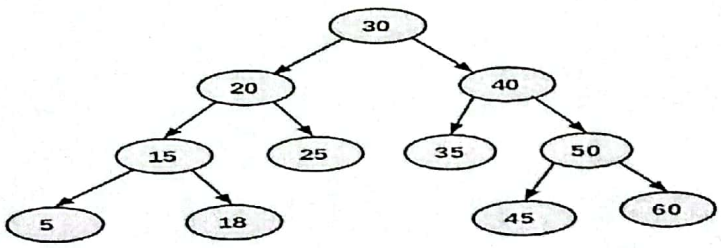
   i. Convert from infix to prefix and postfix: A * B + C / D and (A - B/C) * (A/K-L)

   ii. Convert to postfix notation (3.4+2^(5-e))/(1+5/5) using stack.

7. In computer science, a binary search tree, also called an ordered or sorted binary tree, is a rooted binary tree data structure whose internal nodes each store a key greater than all the keys in the node's left subtree and less than those in its right subtree.

   i. Write down the inorder, preorder and postorder tree traversal of the following graph.

   ii. Your task is to implement to the following tree with insert 2, 1 and delete 18, 45, 40,20 with precise explanation(No need to code)



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